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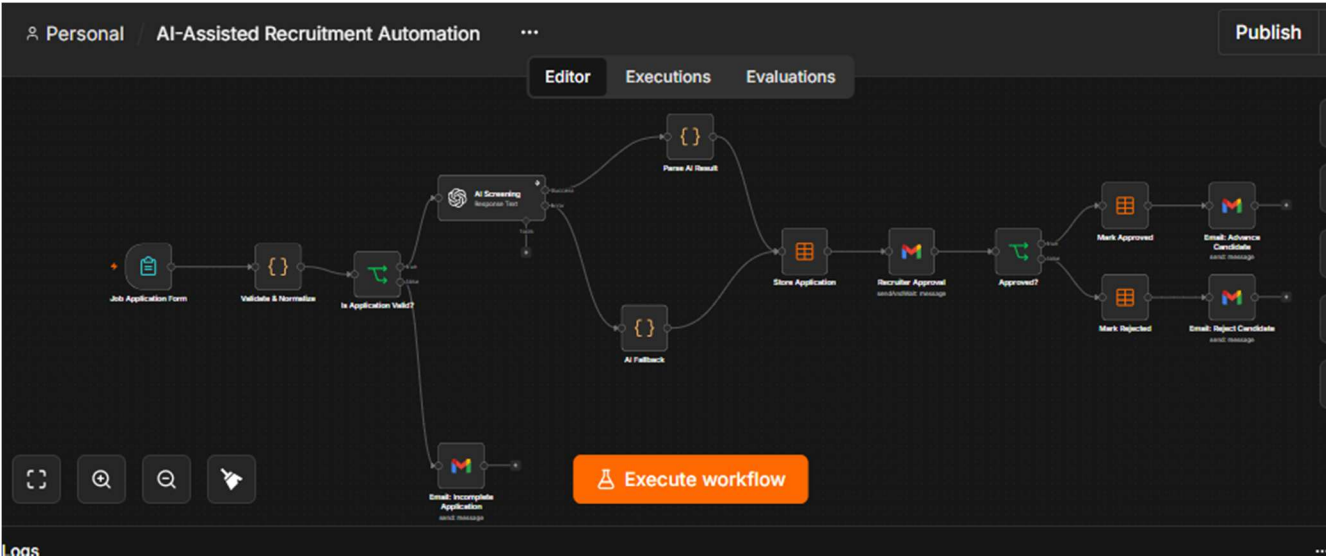
INTERN AT: Petalnex Pvt. Ltd.

INTERNSHIP DOMAIN: AI AUTOMATION

Day 32 — Solution Architecture

End-to-end architecture: Trigger → Validation → Business Rule → AI → Parse/Validate → Fallback → Storage → Human Approval → Status → Notification.

1. Current Workflow Architecture



2. Component Inventory

Group	Node	Purpose
Trigger	Job Application Form	Receive candidate/application data.
Validation	Validate & Normalize	Check required fields and normalize data.
Routing	Application Valid?	Separate valid and incomplete applications.
AI	AI Screening	Perform initial language-based candidate assessment.
AI Output	Parse AI Result	Parse structured AI response.
Resilience	AI Fallback	Handle AI failure in a controlled way.
Storage	Store Application	Persist application/screening data.

Human Review	Recruiter Approval	Capture final recruiter decision.
Decision	Approved?	Route approval/rejection.
Status	Mark Approved	Update approved status.
Status	Mark Rejected	Update rejected status.
Notification	Email Advance Candidate	Notify approved candidate.
Notification	Email Reject Candidate	Notify rejected candidate.
Exception	Email: Incomplete Application	Handle missing required data.

3. Integrations, Models and Data Stores

Technology/Service	Role
n8n	Workflow orchestration and execution.
OpenAI model	AI-assisted screening; use the exact model configured in n8n.
Gmail	Candidate and recruiter/admin notifications.
n8n data table/data store	Application and screening persistence; use the actual configured store.
Job Application Form	Application intake.

4. Credentials and Security

- Store OpenAI and Gmail credentials in n8n credential management.
- Never hard-code API keys, passwords or OAuth tokens in workflow nodes.
- Do not include secrets in screenshots, workflow exports or Git repositories.
- Restrict n8n editor/workflow access to authorized users.
- Minimize candidate data sent to external AI services.
- Use synthetic/test candidate data in presentations and shared artifacts.

5. Failure Handling

Failure	Handling
Missing required data	Application Valid? routes to incomplete path.
AI unavailable	AI Fallback prevents silent failure.
Malformed AI response	Parse AI Result validates/rejects invalid structure.
Storage failure	Execution should surface the failure rather than claim success.
Email failure	Execution error is visible; retry/manual follow-up is possible.
Invalid recruiter decision	Accept only defined decision values.

Credential/authentication failure	Stop affected integration without exposing credential values.
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6. Design Principles

- Build deterministic core before AI.
- Use AI only where reasoning/understanding is needed.
- Validate AI output before downstream use.
- Keep final hiring decision human-controlled.
- Make failures visible and recoverable.
- Keep credentials separate from workflow logic.